

6. [Maximum mark: 8]

A continuous random variable X has probability density function

$$f(x) = \begin{cases} \frac{1}{5} & 0 \leq x < 2 \\ -\frac{x}{30} + \frac{4}{15} & 2 \leq x \leq 8 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find $E(X)$. [3]
- (b) Given that $E(c - 2X) = 0$, where c is a constant, determine the value of c . [2]
- (c) Find the median of X . [3]

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